

Informative Power-System Probing via Sequential Bayesian Experimental Design

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Abstract—Uncertainty in inertia and frequency-response coefficients complicates the characterization of power-system dynamics. We formulate active probing as sequential Bayesian experimental design, selecting pulse durations from noisy measurements while carrying the physical state between probes. Six methods are compared on reduced IEEE 9-bus and IEEE 14-bus swing models under common duration constraints and probe counts. Deep adaptive design (DAD), Step-DAD, and reinforcement-learning-based sequential design (RL-sBOED) obtain higher mean sequential prior-contrastive (sPCE) information scores than optimized fixed-sequence, myopic, and random probing. On IEEE 9-bus, DAD improves the mean score over Fixed by 0.09–0.13 nats. DAD and RL-sBOED select probes through trained policy evaluations; Myopic and Step-DAD additionally require online simulation and optimization. IEEE 14-bus scores approach the finite-contrast estimator ceiling, limiting comparisons among learned methods. The results demonstrate the potential of learned sequential policies for informative probing within the specified benchmark. Code and saved results are available at https://github.com/alexschmidt123/tpec2027_power_grid.

Index Terms—Bayesian experimental design, power-system identification, adaptive probing, expected information gain, rate of change of frequency.

I. INTRODUCTION

As inverter-based resources displace synchronous generation, changing inertia and frequency response increase the need to characterize grid dynamics [1]. Measurement-based inertia estimation uses frequency and active-power responses to disturbances [2]; controlled probing offers an alternative when naturally occurring events do not provide suitable excitation. Field tests on the western North American interconnection demonstrate that probing-signal design affects the quality of electromechanical-mode estimates [3], and recent work applies probing to inertia and frequency-response estimation in systems with inverter-based resources [4].

The design of informative excitation is a central problem in system identification [5]. Here, we study a narrower question: how should successive pulse durations adapt to measurements when each probe begins from the physical state left by the previous probe? Duration changes both the injected energy and the ensuing transient. Because the state is not reset, a decision also changes the conditions under which later

measurements are collected. Selecting a sequence therefore requires accounting for both immediate information and future experiments.

Sequential Bayesian optimal experimental design provides a framework for this tradeoff [6]. A fixed sequence cannot respond to incoming measurements, whereas a myopic design optimizes only the next observation. Deep adaptive design (DAD) [7] and reinforcement-learning-based sequential design (RL-sBOED) [8] learn history-dependent policies using a full-sequence information objective. Step-DAD [9] additionally refines a pretrained policy during the experiment, trading online computation for further adaptation.

We formulate duration selection for reduced IEEE 9-bus and IEEE 14-bus swing models with unknown machine inertia and effective frequency-response coefficients. Our contributions are a carried-state Bayesian probing formulation, a comparison of six methods under a common restricted duration space and probe count, and an analysis of information scores, selected durations, and algorithmic computational requirements.

II. PROBLEM FORMULATION

We seek to reduce uncertainty about grid parameters using active probing within a budget of T design steps. Here $t = 1, \dots, T$ indexes design steps, d denotes elapsed time within each probe window, and D_t is the selected pulse duration.

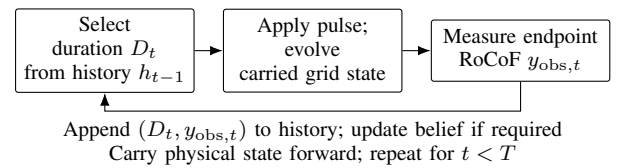


Fig. 1. Sequential probing loop. Adaptive methods use observation history; Fixed and Random select durations without observation feedback.

A. Latent Parameters and Power-System Model

For a reduced network with m machines, the latent parameter vector is $\theta = (M_1, \dots, M_m, K_1, \dots, K_m)$, where M_i and K_i are inertia and effective frequency-response coefficients. The prior $p_0(\theta)$ has independent components $M_i \sim$

TABLE I
PRIOR SCALES AND DAMPING COEFFICIENTS

Network	Bus	$\bar{M}_i$	$\bar{K}_i$	D_i
IEEE 9-bus	1	0.1254141	0.2937500	0.0125414
	2	0.0339531	0.1312500	0.0033953
	3	0.0159685	0.0750000	0.0015969
IEEE 14-bus	1	0.0212207	0.0784314	0.0021221
	2	0.0344836	0.1274510	0.0034484
	3, 6, 8	0.0265257	0.0980392	0.0026526

$\mathcal{U}(0.7\bar{M}_i, 1.3\bar{M}_i)$ and $K_i \sim \mathcal{U}(0.1\bar{K}_i, \bar{K}_i)$. One draw of θ remains fixed throughout all T probes.

Let δ_i be rotor angle, ω_i angular-speed deviation, and u_i the applied probing power. The reduced swing model is

$$\begin{aligned} \dot{\delta}_i &= \omega_i, \\ M_i \dot{\omega}_i &= u_i - \left(D_i + \frac{K_i}{2\pi} \right) \omega_i \\ &\quad - \sum_{j=1}^m B_{ij} \sin(\delta_i - \delta_j). \end{aligned} \quad (1)$$

Dots denote derivatives with respect to physical time d . In (1), M_i scales acceleration, while K_i contributes a restoring power response proportional to frequency deviation. It is an effective coefficient, not an explicit turbine–governor model. The damping D_i and network couplings B_{ij} are known; frequency deviation is $\Delta f_i = \omega_i/(2\pi)$ in Hz. The units of M_i , K_i and D_i are pu s²/rad, pu/Hz and pu s/rad, respectively, on a 100-MVA base at nominal frequency 60 Hz. All angles and speed deviations start at zero. Net mechanical injections are zero in this deviation model, making the initial unforced state an equilibrium.

The reduced IEEE 9-bus model retains buses 1, 2, and 3; the reduced IEEE 14-bus model retains buses 1, 2, 3, 6, and 8. Test-case reactances [10] define a lossless Laplacian $\mathbf{L}$. Eliminating algebraic buses l while retaining machines g gives $\mathbf{L}_{\text{red}} = \mathbf{L}_{gg} - \mathbf{L}_{gl}\mathbf{L}_{ll}^{-1}\mathbf{L}_{lg}$, with $B_{ij} = -(\mathbf{L}_{\text{red}})_{ij}$ for $i \neq j$ and $B_{ii} = 0$. Resistance, voltage dynamics, and transformer tap effects are omitted. Table I summarizes the model parameterization. Nominal inertia is computed as $\bar{M}_i = 2H_i/(2\pi f_0)$, with $f_0 = 60$ Hz. For IEEE 9-bus, $H = (23.64, 6.40, 3.01)$ s matches published machine data [11, Table 2]. For IEEE 14-bus, $H = (4, 6.5, 5, 5, 5)$ s is prescribed at buses (1, 2, 3, 6, 8) as assumed benchmark values. For both networks, $\bar{K}_i = K_{\text{tot}}w_i$, with $K_{\text{tot}} = 0.5$ pu/Hz. The allocation weights are $w = (0.5875, 0.2625, 0.15)$ for IEEE 9-bus and $w = (8, 13, 10, 10, 10)/51$ for IEEE 14-bus. Fixed damping satisfies $D_i = (0.1 \text{ s}^{-1})\bar{M}_i$. The response scales, damping values, and prior ranges are benchmark assumptions rather than validated operational grid parameters.

B. Probe Design and Observation Model

The probe design is $\xi_t = (A_t, B_t, D_t)$: amplitude, bus location, and probe signal duration. This study fixes $A_t = 0.05$

pu and $B_t = 1$; only D_t is optimized. The raised-cosine input for $0 \leq d \leq D_t$ is

$$u_{B_t,t}(d) = \frac{A_t}{2} [1 - \cos(2\pi d/D_t)]. \quad (2)$$

The input is zero for $D_t < d \leq D_{\text{obs}}$ and at all other buses. Fixing amplitude and location isolates duration selection rather than joint actuator design. The pulse injects energy proportional to $A_t D_t/2$; equal probe counts therefore do not imply equal injected energy. The observation window is $D_{\text{obs}} = 3.5$ s. The full angle and speed state at its end becomes the initial state of the next window, without resetting the system. Consequently, the response depends on the complete design sequence $\xi_{1:t} = (\xi_1, \dots, \xi_t)$.

Write $\Delta f_{1,t}(d; \theta, \xi_{1:t})$ for the resulting bus-1 frequency deviation. The noise-free response is the signed endpoint rate of change of frequency (RoCoF), denoted by y_t :

$$y_t(\theta, \xi_{1:t}) = \frac{\Delta f_{1,t}(D_{\text{obs}}) - \Delta f_{1,t}(D_{\text{obs}} - \Delta d_{\text{samp}})}{\Delta d_{\text{samp}}}. \quad (3)$$

The frequency arguments $\theta, \xi_{1:t}$ are suppressed in the numerator. With sampling interval $\Delta d_{\text{samp}} = 0.025$ s, the measured response follows

$$y_{\text{obs},t} = y_t(\theta, \xi_{1:t}) + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma^2), \quad (4)$$

where $\sigma = 0.005$ Hz/s and noise is independent across windows. The endpoint measurement summarizes the post-pulse transient, rather than the initial inertial response. A single fixed sensor keeps the observation model simple and identical across methods, but discards information in the full frequency trajectory and at other buses. This is a benchmark choice, not an optimized sensor configuration. Each observation becomes available at window end. Decision latency is idealized as zero in simulated time.

C. Sequential Decisions and Bayesian Updating

The data after step t are the ordered design–observation history $h_t = ((\xi_k, y_{\text{obs},k}))_{k=1}^t$, with $h_0 = \emptyset$. A policy π selects ξ_t using only observed history h_{t-1} , with amplitude and bus fixed as above. It has no access to true parameters or physical state. Feasible durations satisfy

$$\begin{aligned} 0.2 &\leq D_t \leq 3, \quad t = 1, \dots, T, \\ D_t - D_{t-1} &\geq \Delta D_{\text{min}}, \quad t = 2, \dots, T, \end{aligned} \quad (5)$$

The increasing-duration restriction defines the benchmark design space; it is not a physical requirement of the swing equations. Applying it to every method permits comparisons within this space, but excludes nonmonotonic sequences that may be more informative. Thus, optimality in (8) is restricted to Π . Here, $\Delta D_{\text{min}} = 0.01$ s. At step t , the upper bound is $3 - (T - t)\Delta D_{\text{min}}$ to leave enough room for the remaining probes. The lower bound starts at 0.2 s and then becomes $D_{t-1} + \Delta D_{\text{min}}$. All methods map a unit action to this interval without sorting or discretizing durations. Let Π denote the set of policies that satisfy these constraints. Any auxiliary random numbers used by a policy are independent of θ ; its actions depend on the observed history.

The Gaussian likelihood $p(y_{\text{obs},t} \mid \theta, h_{t-1}, \xi_t)$ has mean $y_t(\theta, \xi_{1:t})$ and variance σ^2 . Bayes' rule updates parameter uncertainty after each measurement:

$$p(\theta \mid h_t) = \frac{p(y_{\text{obs},t} \mid \theta, h_{t-1}, \xi_t) p(\theta \mid h_{t-1})}{p(y_{\text{obs},t} \mid h_{t-1}, \xi_t)}. \quad (6)$$

The denominator integrates the numerator over θ , with $p(\theta \mid h_0) = p_0(\theta)$. Each candidate parameter vector propagates its own physical state under the observed designs. Since the policy uses only observed history and independent auxiliary randomness, its action probabilities cancel from the Bayesian update.

D. Expected Information Gain and Design Objective

The expected information gain (EIG) of policy π measures the mean Kullback–Leibler divergence between the terminal posterior and the prior:

$$\mathcal{I}_T(\pi) = \mathbb{E}_{h_T \mid \pi} [\mathcal{D}_{\text{KL}}(p(\theta \mid h_T) \parallel p_0(\theta))]. \quad (7)$$

Here $\mathcal{D}_{\text{KL}}(p \parallel q) = \int p(\theta) \log[p(\theta)/q(\theta)] d\theta$; information gain is measured in nats. The expectation averages over parameter draws, measurement noise, and any policy randomness. The sequential Bayesian design problem is

$$\pi^* \in \arg \max_{\pi \in \Pi} \mathcal{I}_T(\pi). \quad (8)$$

Equation (8) defines the common information objective; the methods below differ in how they approximate or forgo its optimization. All receive the same number of probes. Computation is assessed separately.

Evaluating EIG requires expensive marginal-likelihood calculations. We therefore use the sequential prior-contrastive (sPCE) lower bound [7] for training and evaluation, instead of the true EIG. Let $\theta^{(0)} \sim p_0$ generate the observed history and let $\theta^{(1)}, \dots, \theta^{(L)}$ be independent contrastive samples from the prior. For any candidate θ , define the observation likelihood along the realized sequence as

$$\mathcal{L}_t(\theta) = \prod_{k=1}^t p(y_{\text{obs},k} \mid \theta, h_{k-1}, \xi_k). \quad (9)$$

The sampled information score is then

$$\hat{I}_t = \log \frac{\mathcal{L}_t(\theta^{(0)})}{\frac{1}{L+1} \sum_{\ell=0}^L \mathcal{L}_t(\theta^{(\ell)})}. \quad (10)$$

Likelihoods are evaluated in the log domain. Averaging $\hat{I}_T$ estimates the sPCE lower bound on $\mathcal{I}_T(\pi)$, not the exact EIG. Individual scores may be negative but are bounded above by $\log(L+1)$. With $L = 128$, the ceiling is 4.8598 nats; near-ceiling scores can conceal differences in EIG. We report these estimated lower-bound values as information scores throughout the results. The policy does not receive the generating parameters.

III. COMPARED DESIGN METHODS

All six methods use the same observation model and feasible duration interval (5). Adaptation and planning horizon are distinct: an adaptive method can be myopic, whereas a fixed sequence can be optimized jointly for the full horizon. The algorithms are adaptations of existing methods to the carried-state grid simulator.

A. Nonadaptive Designs: Random and Fixed

Random samples $z_t \sim \mathcal{U}(0, 1)$ independently at each step and sets $D_t = D_t^{\min} + (D_t^{\max} - D_t^{\min})z_t$. Here $D_1^{\min} = 0.2$, $D_t^{\min} = D_{t-1} + \Delta D_{\min}$ for $t > 1$, and $D_t^{\max} = 3 - (T-t)\Delta D_{\min}$. Thus, random samples are uniform within the remaining feasible interval and are not uniform over feasible complete sequences. Random does not use observations and needs no offline training.

Fixed is a nonadaptive Bayesian design: all durations are optimized jointly before observations are collected [6, Sec. 2.4]. The ordered sequence is optimized using the full-sequence sPCE objective. It compares short, central, long, and spread interior initializations using validation, then optimizes the selected initialization. Each training seed yields a separately fitted sequence. During evaluation, Fixed executes that sequence without adapting to observations.

B. Online One-Step Design: Myopic

Unlike Fixed, Myopic uses observation feedback. It follows one-step adaptive BOED, maximizing the expected information from the next observation under the current posterior [6, Sec. 2.4]. Our implementation maintains a weighted particle approximation to this posterior. Before each decision after the first, particles propagate their own states under the preceding duration and update likelihood weights using the latest observation. Continuous search then maximizes expected information from the next probe only, using 16 proposals, three search rounds, and 32 simulated observations per proposal. Candidate durations obey the same constraints as all other methods. No offline policy is trained.

C. Offline-Trained Adaptive Methods: DAD and RL-sBOED

DAD and RL-sBOED use neural policies trained on simulated sequences drawn from the same parameter prior as Fixed. Both methods receive the stage index and ordered trajectories, including past durations, scaled observations, and availability masks; future slots are masked. During evaluation, actions are generated via a forward pass without online optimization or explicit particle-posterior updates.

DAD [7] employs a deterministic two-layer, 64-unit tanh network, a trainable stage bias, and a sigmoid action map. Offline training maximizes sPCE by differentiating through the entire simulated sequence, including both observations and carried states. In joint Fixed/DAD training, validation determines whether to use random or interior Fixed-based initialization. Section V-C addresses the associated initialization cost.

RL-sBOED [8] uses a REDQ-style ensemble actor-critic framework with ten critics and five replay updates per trajectory batch. The actor consists of two 128-unit ReLU layers. Per-step information rewards are defined as $R_t = \hat{I}_t - \hat{I}_{t-1}$, with $\hat{I}_0 = 0$ and a discount factor $\gamma = 1$. As a result, the cumulative reward equals the terminal sPCE score. Offline training accounts for future design steps; online evaluation uses the trained actor without additional planning simulations. Entropy is regularized by normalized tanh actions and a target entropy of -1 ; reported information scores do not include entropy bonuses. We use the actor’s deterministic output for evaluation.

D. Online Refinement: Step-DAD

Step-DAD [9] starts each episode from its selected DAD checkpoint and leaves the first probe unchanged. Before subsequent decisions, a 128-particle posterior supplies candidate parameters and their carried states for simulations of the remaining sequence. A fresh Adam optimizer performs four updates per decision (batch size 32, learning rate 10^{-3}). A fixed conditional validation batch selects the best candidate, including the unchanged policy. Retained weights persist within an episode only. The total refinement budget is $4(T - 1)$ updates. Myopic and Step-DAD require posterior assimilation only before subsequent decisions; terminal information scoring is separate.

IV. EXPERIMENTAL SETTINGS

A. Simulation and Training Protocol

We evaluate horizons $T = 3, 4, 5$, with one scalar observation per probe. Each network has separately trained policies; sensor settings and duration constraints are shared.

The fourth-order Runge-Kutta (RK4) method is used to integrate the swing equations, with a step size of $1/640$ s. Gradients propagate through the carried states using numerical Jacobians: state perturbations are $10^{-5} \max(1, |x_j|)$ for state coordinate x_j , and duration perturbations are 10^{-5} s, clipped at the bounds. The validation suite compares the policy gradients and the conditional refinement gradients with those obtained by end-to-end finite differences.

Offline training uses 2000 updates, a batch size of 32, and three independent training seeds (101, 202, and 303). The DAD and Fixed learning rates are both 10^{-3} . Validation is carried out using 128 systems every 100 updates. For the sPCE objective, 128 contrastive samples are used in the training, validation, and evaluation phases. The Myopic and Step-DAD methods each use 128 particles for online posterior planning, and Step-DAD carries out four refinement updates after each observation so long as design steps remain. The evaluation comprises seeds 1001, 1002, and 1003, with 512 simulated experimental sequences per evaluation seed. Validation cases are not used for gradient updates. Each trained policy is assessed over 1536 episodes. The draws for parameters and measurement noise are paired across the different methods, whereas the online planner makes use of its own separate random stream. Since the Random and Myopic methods show

no dependence on the training seed under these streams, identical evaluations can be obtained for all training seeds. Fixed is optimized independently for each training seed.

B. Evaluation Protocol

For trained methods, we average the evaluation-seed means within each training seed and report the mean $\pm$ sample standard deviation across the three training seeds. For Random and Myopic, we report the mean $\pm$ sample standard deviation across evaluation-seed means rather than episode-level dispersion. Training, checkpoint validation, and testing use separate random seeds.

All offline and online timing measurements use NVIDIA A100 PCIe 40-GB GPUs; their scope is specified in Section V-C.

V. RESULTS AND DISCUSSION

A. Information Acquired at Equal Probe Budgets

Tables II and III report mean sPCE information scores in nats, with the seed-level standard deviations defined above. All three learned adaptive methods have higher mean sPCE scores than Random, Fixed, and Myopic in the reported network-horizon comparisons.

TABLE II
IEEE 9-BUS INFORMATION SCORES (SPCE, NATS). MEAN $\pm$ SD ACROSS TRAINING-SEED MEANS; RANDOM/MYOPIC USE EVALUATION-SEED MEANS.

Method	$T = 3$	$T = 4$	$T = 5$
Random	0.4321 ± 0.0310	0.4924 ± 0.0342	0.5392 ± 0.0471
Fixed	2.0112 ± 0.0116	2.3192 ± 0.0004	2.5066 ± 0.0013
Myopic	1.9736 ± 0.0438	2.1707 ± 0.0366	2.2407 ± 0.0547
DAD	2.1366 ± 0.0161	2.4095 ± 0.0080	2.6351 ± 0.0179
RL-sBOED	2.1257 ± 0.0027	2.3963 ± 0.0333	2.5307 ± 0.0114
Step-DAD	2.1471 ± 0.0110	2.4382 ± 0.0048	2.6621 ± 0.0145

TABLE III
IEEE 14-BUS INFORMATION SCORES (SPCE, NATS). UNCERTAINTY FOLLOWS TABLE II; THE ESTIMATOR CEILING IS 4.8598 NATS.

Method	$T = 3$	$T = 4$	$T = 5$
Random	0.7541 ± 0.0047	0.8360 ± 0.0297	0.8464 ± 0.0252
Fixed	4.2623 ± 0.0402	4.4946 ± 0.0345	4.6624 ± 0.0083
Myopic	4.2907 ± 0.0368	4.3990 ± 0.0177	4.4641 ± 0.0280
DAD	4.7576 ± 0.0362	4.8325 ± 0.0047	4.8443 ± 0.0026
RL-sBOED	4.7445 ± 0.0022	4.8275 ± 0.0112	4.8383 ± 0.0108
Step-DAD	4.7527 ± 0.0275	4.8297 ± 0.0067	4.8446 ± 0.0042

On IEEE 9-bus, DAD exceeds Fixed by 0.1253, 0.0903, and 0.1286 nats at $T = 3, 4, 5$, respectively. Step-DAD improves the DAD mean by 0.0105, 0.0287, and 0.0269 nats. The table SDs describe separate method variability, not confidence intervals for paired differences; these gains are therefore reported descriptively. The IEEE 9-bus information scores are well below the estimator ceiling.

On IEEE 14-bus, learned-policy scores approach the 4.8598-nat ceiling, preventing reliable EIG rankings. Step-DAD changes the DAD mean by -0.0049 , -0.0028 , and $+0.0003$ nats at $T = 3, 4, 5$. Fixed exceeds Myopic at $T = 4, 5$ under the stated search budget; a larger Myopic budget could change this comparison.

B. Selected Probe Durations

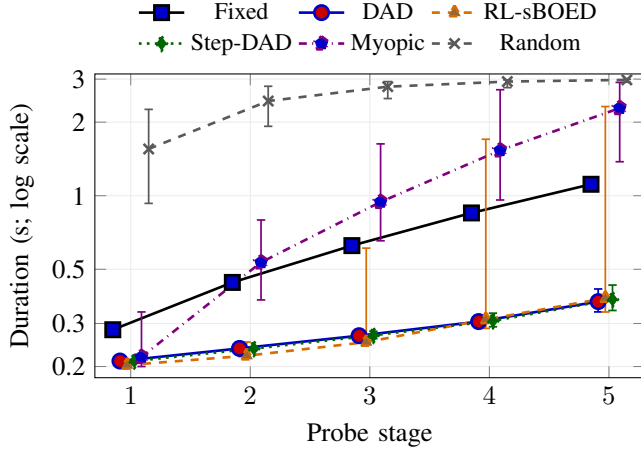


Fig. 2. IEEE 14-bus pulse durations at $T = 5$: medians and interquartile ranges for all six methods. Fixed and learned methods: 4608 histories over three seeds; Myopic and Random: 1536 each. Horizontal offsets separate methods. Bars span the pooled 25th–75th percentiles across episodes and training seeds; they are not confidence intervals. Ordering is imposed.

At $T = 5$, median final durations are 0.369 s for DAD, 0.376 s for Step-DAD, 1.114 s for Fixed, 2.283 s for Myopic, and 2.979 s for Random (Fig. 2). Random’s sequential interval sampling concentrates later pulses near the upper bound. The methods therefore select different duration and energy allocations. RL-sBOED varies substantially across fitted policies: final-duration medians are 2.585, 0.381, and 0.324 s for training seeds 101, 202, and 303, respectively. This between-seed variation contributes to its wide pooled interquartile ranges. Ordering is imposed; unrestricted and energy-matched comparisons remain open.

C. Computational Cost

Tables IV and V report costs for training seed 101 and test seed 1001. Offline training uses NVIDIA A100 PCIe 40-GB GPUs and includes initialization, optimization, and checkpoint validation. DAD’s entry sums the matched Fixed and DAD training stages; Step-DAD inherits this preparation. Online cost is the mean sum of per-stage decision timers over 512 episodes per method, including posterior assimilation and refinement but excluding true-system simulation and information scoring. All timings use NVIDIA A100 PCIe 40-GB GPUs. IEEE 9-bus offline runs used a Xeon Platinum 8352Y CPU; IEEE 9-bus online runs and IEEE 14-bus runs used a Xeon Gold 6248R CPU. A separate warm-up episode per method is discarded before the IEEE 9-bus timing evaluation. These

are descriptive measurements, not a controlled cross-network hardware comparison.

Let C_s denote one candidate-window simulation cost and C_π a policy forward pass. Dense m -machine RK4 integration over N time steps gives $C_s = O(Nm^2)$.

Online policy work is $O(T)$ for Random/Fixed and $O(TC_\pi)$ for DAD/RL-sBOED. Myopic uses $O(TQC_e + TPC_s)$, where $P = 128$ particles, $Q = 48$ candidate evaluations per decision, and C_e includes candidate simulation and likelihood calculations. Step-DAD refinement simulates $UB(L+1)T(T-1)/2$ candidate windows before accounting for numerical Jacobians and conditional validation, with $U = 4$, $B = 32$, and $L = 128$. Posterior assimilation adds $O(TPC_s)$ work.

TABLE IV
IEEE 9-BUS RECORDED COMPUTATIONAL COSTS

Method	$T = 3$	$T = 4$	$T = 5$
Offline preparation (hours)			
Random	0	0	0
Fixed	0.554	0.802	1.063
DAD	1.116	1.608	2.139
RL-sBOED	0.100	0.117	0.129
Myopic	0	0	0
Step-DAD	1.116	1.608	2.139
Online decisions (ms per sequence)			
Random	0.167	0.229	0.303
Fixed	0.849	1.135	1.472
DAD	1.012	1.345	1.745
RL-sBOED	1.147	1.781	7.692
Myopic	4418	5952	7514
Step-DAD	3083	7689	14346

TABLE V
IEEE 14-BUS RECORDED COMPUTATIONAL COSTS

Method	$T = 3$	$T = 4$	$T = 5$
Offline preparation (hours)			
Random	0	0	0
Fixed	1.880	2.750	3.720
DAD	3.810	5.457	7.418
RL-sBOED	0.163	0.205	0.249
Myopic	0	0	0
Step-DAD	3.810	5.457	7.418
Online decisions (ms per sequence)			
Random	0.158	0.207	0.285
Fixed	1.141	1.906	1.952
DAD	1.440	1.513	1.733
RL-sBOED	1.413	1.550	1.905
Myopic	9998	13433	16918
Step-DAD	9245	23320	45089

Let C_g include per-window simulation and gradient work. Offline training with J updates, batch size B , and L contrasts requires $O(JB(L+1)TC_g)$ simulation/gradient work for Fixed and DAD. DAD’s complete preparation includes

Reserved for IEEE 14-bus
larger- L evaluation results
 $T = 3, 4, 5$; $L = 10^4, 10^5$

the Fixed stage: $W_F + W_D$; Step-DAD inherits both. Random and Myopic need no offline training. RL-sBOED uses $O(JB(L+1)TC_s)$ simulated trajectory work plus actor-critic replay updates and validation. Offline training times are single-run observations, not seed-averaged estimates. At IEEE 9-bus $T = 5$, Step-DAD adds 0.0269 nats to the DAD mean, while recorded online decision time increases from 1.745 ms to 14.346 s per sequence.

Rescoring n histories costs $O(n(L+1)TC_s)$ simulation work plus $O(nLT)$ likelihood arithmetic, without retraining. These timings describe the recorded implementation and hardware; they do not establish real-time feasibility.

D. Estimator Sensitivity

The ceiling $\log(L+1)$ limits sPCE, not EIG. A mean below it can conceal saturated histories. Larger- L evaluation of the same saved histories is needed to assess this bias without retraining.

E. Scope and Limitations

The findings concern reduced lossless dynamics, the specified priors, monotonic durations, and a fixed actuator and scalar endpoint sensor. IEEE 14-bus has ten unknown parameters but only three to five observations. Information gain measures joint uncertainty reduction; it does not establish accurate identification of each inertia or response coefficient. Training and testing draw from the same prior, so robustness to prior shift, sensor bias, and unmodelled dynamics remains untested. Finite-contrast scoring, particle inference, and ODE integration introduce distinct approximations. In particular, scores below the sPCE ceiling can still include saturated histories. Finally, the simulator carries physical states between probes but idealizes decision latency as zero. Recorded computational costs therefore do not demonstrate real-time deployment or improved control safety.

VI. CONCLUSION

Learned adaptive policies use observation history to select probes over the experimental sequence. When three to five

probes are used on the reduced IEEE 9-bus and IEEE 14-bus models, DAD, RL-sBOED, and Step-DAD obtain higher mean sPCE information scores than Random, Fixed, and Myopic. Estimator saturation limits comparisons among learned policies on the IEEE 14-bus model. DAD and RL-sBOED replace online planning with policy forward passes. Step-DAD adds simulation and gradient refinement, with modest mean score gains on IEEE 9-bus and no consistent improvement on IEEE 14-bus under the original estimator. Further work should examine model mismatch, alternative sensing and design constraints, and decision latency.

REFERENCES

- [1] P. Denholm, T. Mai, R. W. Kenyon, B. Kroposki, and M. O'Malley, "Inertia and the power grid: A guide without the spin," National Renewable Energy Laboratory, Tech. Rep. NREL/TP-6A20-73856, 2020. [Online]. Available: <https://docs.nrel.gov/docs/fy20osti/73856.pdf>
- [2] P. Wall and V. Terzija, "Simultaneous estimation of the time of disturbance and inertia in power systems," *IEEE Transactions on Power Delivery*, vol. 29, no. 4, pp. 2018–2031, 2014.
- [3] J. W. Pierre, N. Zhou, F. K. Tuffner, J. F. Hauer, D. J. Trudnowski, and W. A. Mittelstadt, "Probing signal design for power system identification," *IEEE Transactions on Power Systems*, vol. 25, no. 2, pp. 835–843, 2010.
- [4] J. Peng, J. Tan, P. Koralewicz, E. Mendiola, S. Dong, A. Hoke, and K. Horowitz, "Probing signal-based inertia and frequency response estimation for power systems with high levels of inverter-based resources," in *2024 IEEE Power & Energy Society General Meeting (PESGM)*, 2024, pp. 1–5.
- [5] H. Hjalmarsson, "System identification of complex and structured systems," *European Journal of Control*, vol. 15, no. 3–4, pp. 275–310, 2009.
- [6] X. Huan and Y. M. Marzouk, "Sequential Bayesian optimal experimental design via approximate dynamic programming," arXiv:1604.08320, 2016. [Online]. Available: <https://arxiv.org/abs/1604.08320>
- [7] A. Foster, D. R. Ivanova, I. Malik, and T. Rainforth, "Deep adaptive design: Amortizing sequential Bayesian experimental design," in *Proceedings of the 38th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 139, 2021, pp. 3384–3395.
- [8] T. Blau, E. V. Bonilla, I. Chades, and A. Dezfouli, "Optimizing sequential experimental design with deep reinforcement learning," in *Proceedings of the 39th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 162, 2022, pp. 2107–2128.
- [9] M. Hedman, D. R. Ivanova, C. Guan, and T. Rainforth, "Step-DAD: Semi-amortized policy-based Bayesian experimental design," in *Proceedings of the 42nd International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 267, 2025, pp. 22 904–22 923. [Online]. Available: <https://proceedings.mlr.press/v267/hedman25a.html>
- [10] R. D. Zimmerman, C. E. Murillo-Sanchez, and R. J. Thomas, "MATPOWER: Steady-state operations, planning, and analysis tools for power systems research and education," *IEEE Transactions on Power Systems*, vol. 26, no. 1, pp. 12–19, 2011.
- [11] R. Kaur and D. Kumar, "Transient stability improvement of IEEE 9 bus system using power world simulator," *MATEC Web of Conferences*, vol. 57, p. 01026, 2016.